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INTERNATIONAL UNIVERSITY
جامعة العلمين الدولية

Upnormal behaviour detection

CSE/AIE Graduation Project Presentation

Supervised by : Dr Essam Abdellatef

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Team Members



Ahmed Mostafa
Software Engineer



Mohamed Ahmed
Software Engineer
Team Lead



Abdullah Abdalgawad
AI Engineer



Khalid Abdelsalam
Software Engineer



Omnia Adel
AI Engineer



Tasneem Ibrahim
Ai Engineer



Mariam Elrafei
Software Engineer

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Introduction

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Problem Statement

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Motivation

Objectives

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Requirements

Existing Solutions

Literature Review

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Proposed Solution

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Methodology

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System Architecture

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System Workflow

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UML Diagrams

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Technology Stack

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Dataset

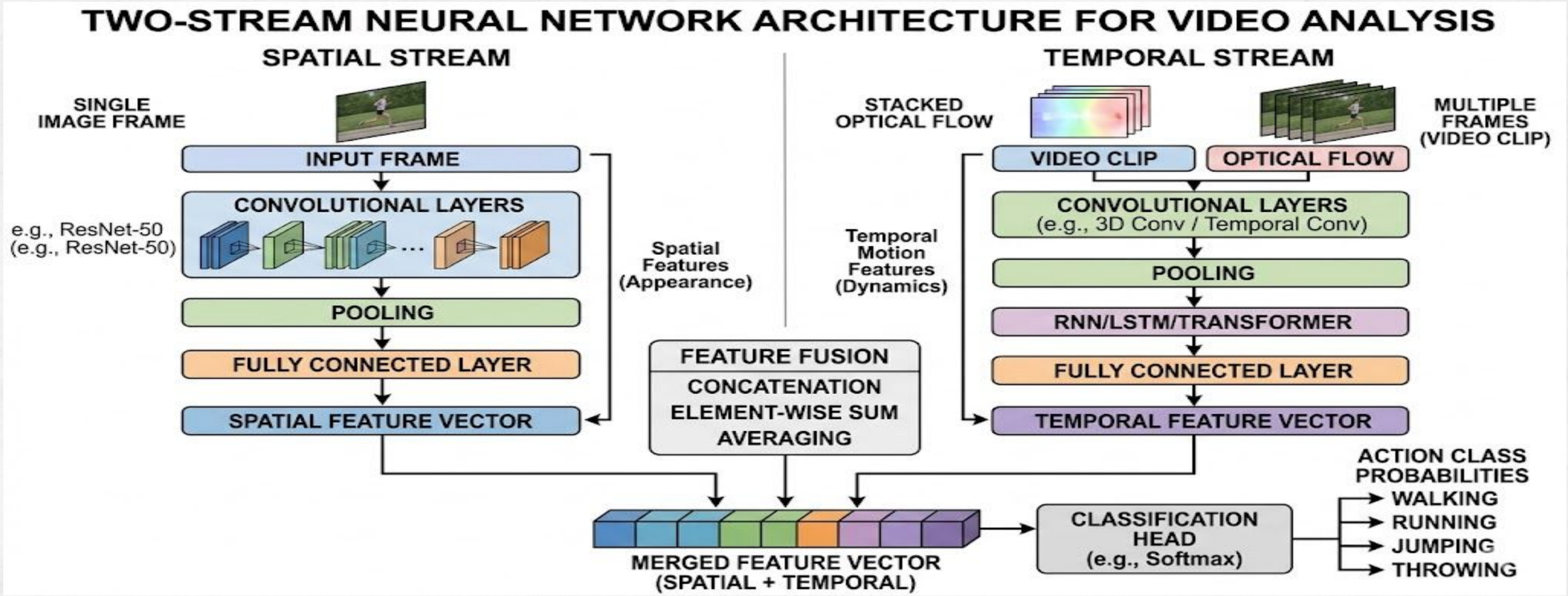
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Data Preprocessing

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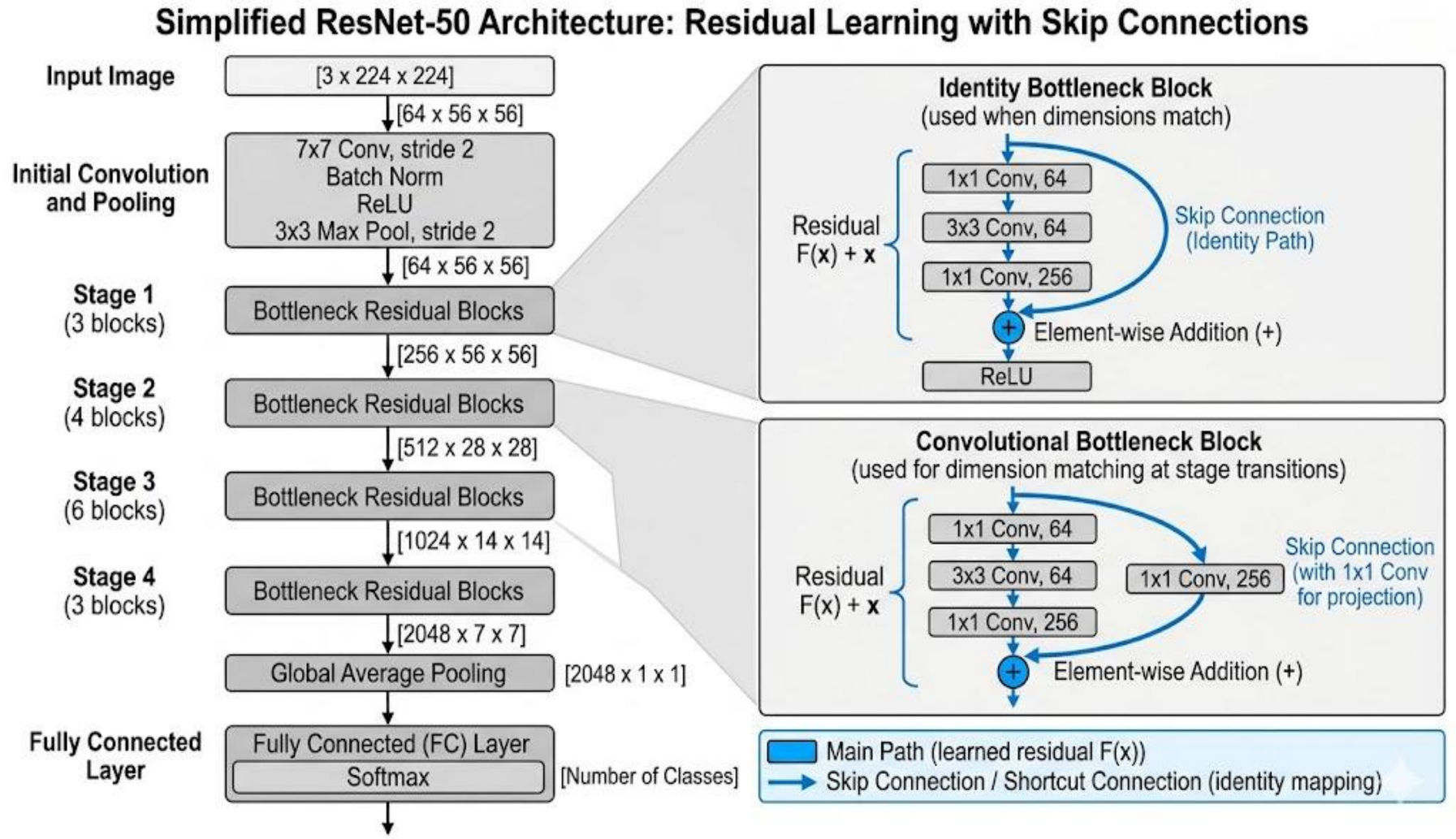
Model Architecture

Two-Stream Architecture



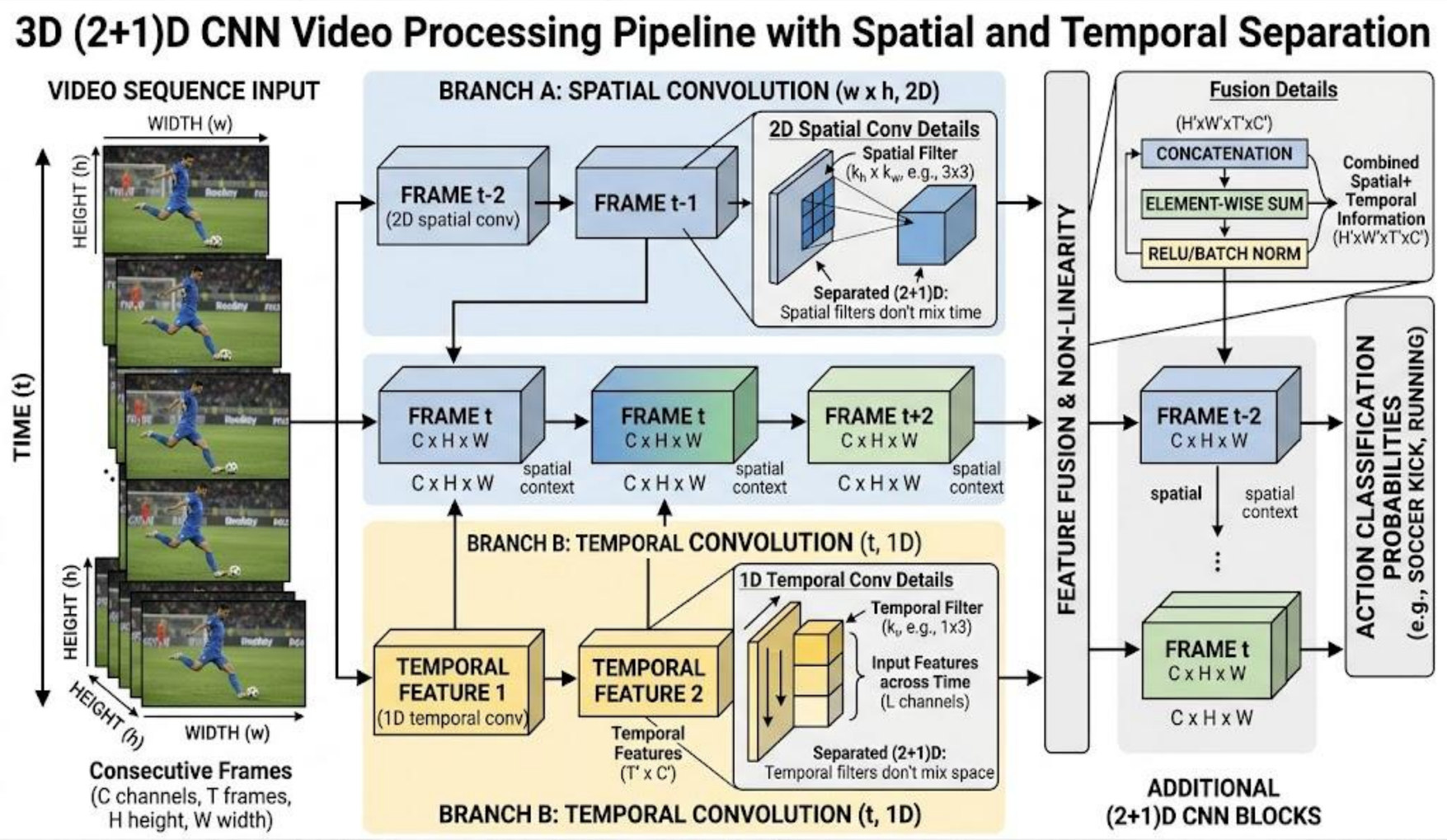
- **Formulation:**
  - ➔ Ffinal=Fuse(Fs,Ft)

Spatial Stream (ResNet-50)



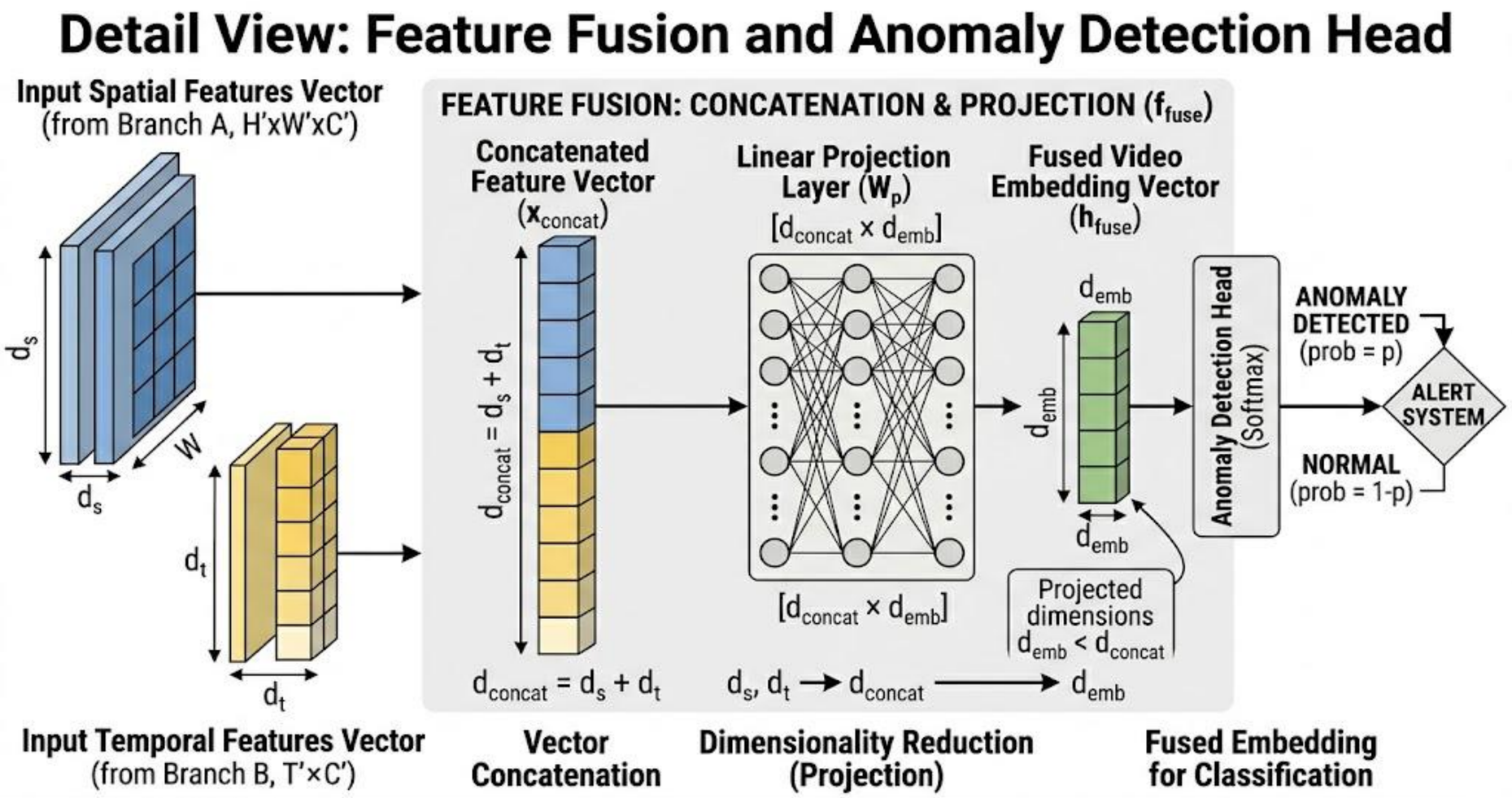
- **Spatial convolution:**
  - ➔ **Fs1 = f (Ws ∗ xt + bs)**
- **Residual Learning:**
  - ➔ **Fs = F (Fs1 + xt)**
- **Final Spatial Output:**
  - ➔ **Ys = f (Ws ∗ Xt + bs) + Xt**

Temporal Stream (R(2+1)D)



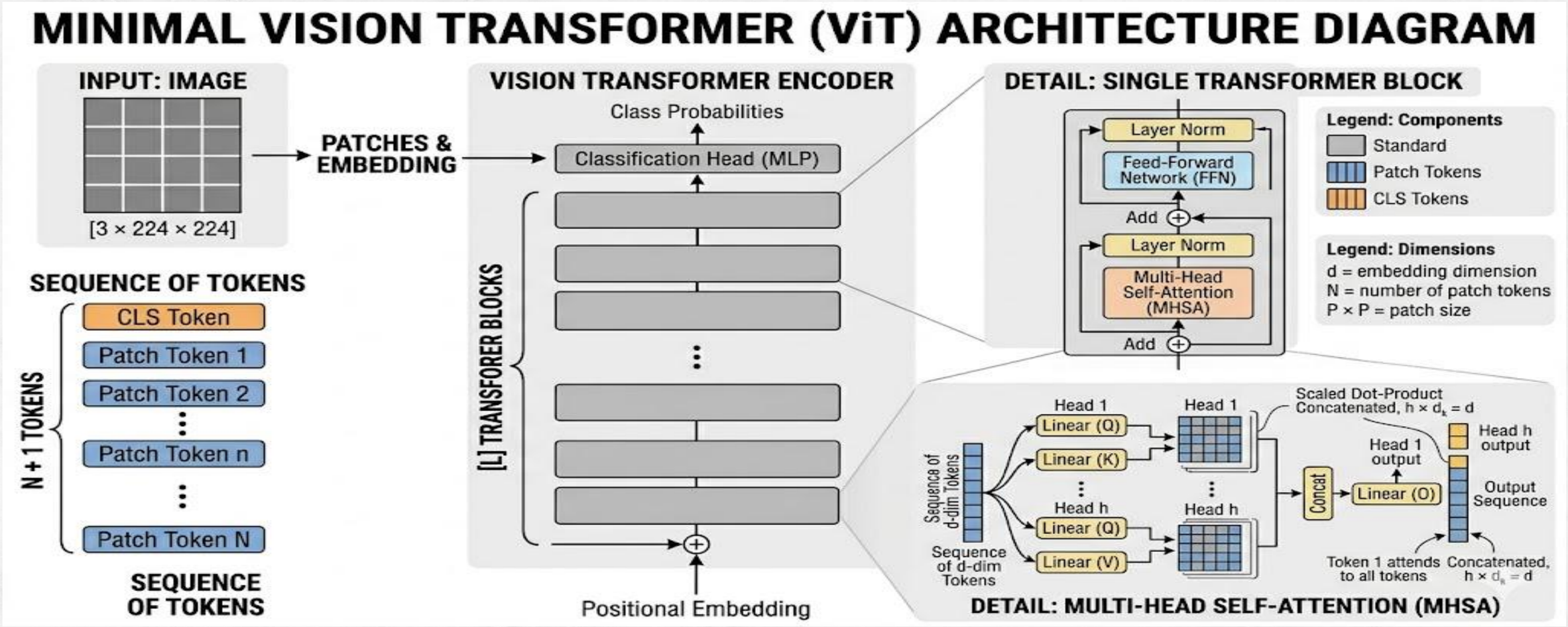
- **Spatial convolution:**
  - ➔ st= fs (Ws ∗ xt)
- **Temporal convolution:**
  - ➔ Ft = ft (Wt * Fst + bt)
- **Residual Learning:**
  - ➔ Yt = F (Fs1 + xt)
- **Final Spatial tOutput:**
  - ➔ Ys = ft (fs (Ws ∗ X)) + X

Feature Fusion



- **Equation (concatenation):**
  - ➔ F = [Fs;Ft]
- **Then projection:**
  - ➔ F′ = WF + b

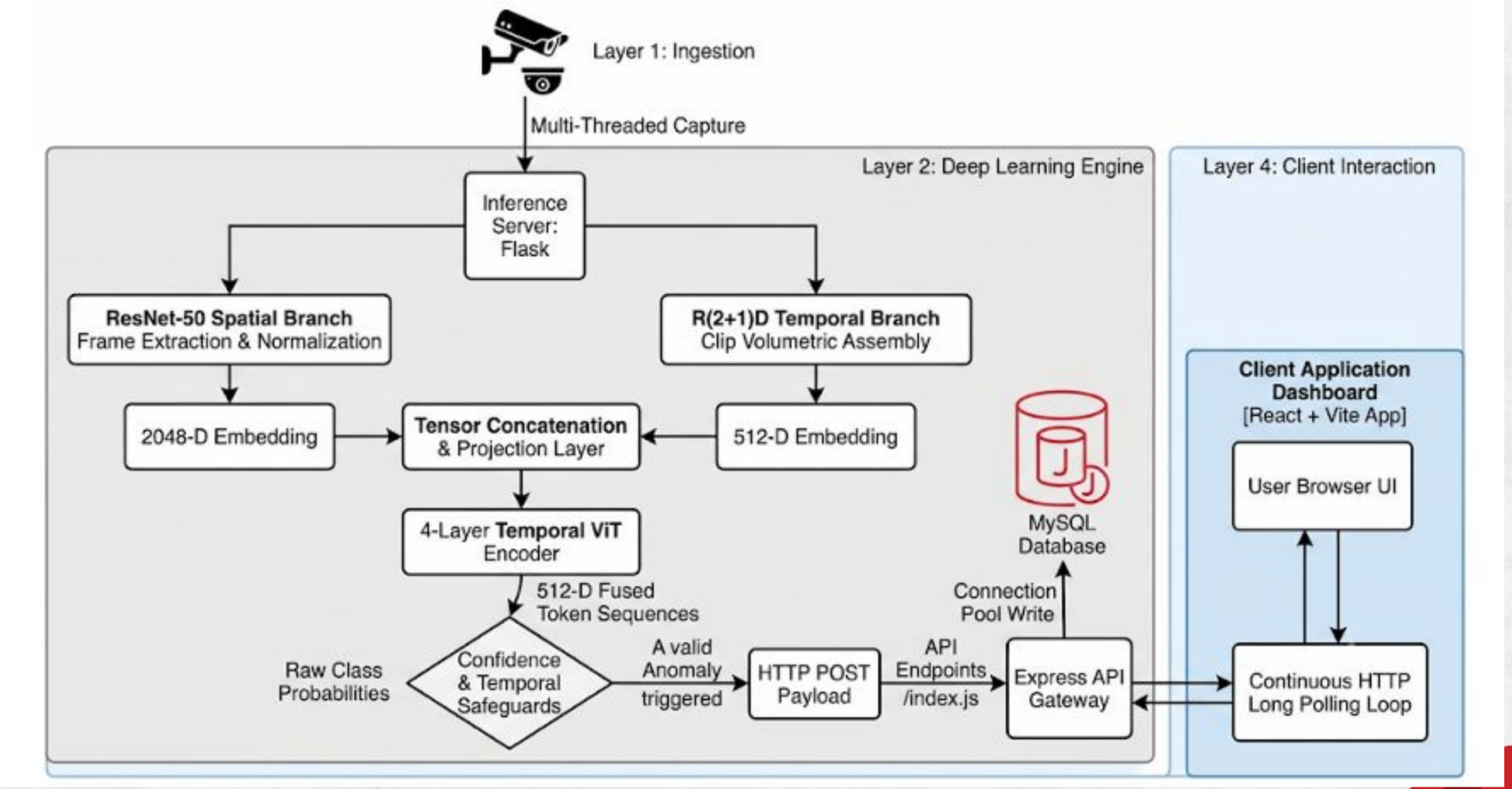
Vision Transformer (ViT)



- Attention(Q,K,V)=softmax{Q*K(Transpose) / square root (d)}*V

Training Process

Implementation



Implementation

Web System Architecture

LIVE MODE

Integrated Data Flow & Backend API Gateway

Data Ingestion

- RTSP Stream Capture
- Multi-threaded Pipeline
- Frame Downsampling

SOURCE
IP Cameras (Edge)

Backend API Gateway

Routing Logic	Protocol
Async Dispatcher	**HTTP/WS**
Auth	**Load**
JWT Tokens	**Balanced**

Manages request queuing and response caching

System Integration

- Alert Dispatch Hub
- Service Orchestration
- Pub/Sub Messaging

TARGET
Responders / DB

Implementation

CHAPTER 5: IMPLEMENTATION DETAILS

AI Pipeline & Optimization

LATENCY 42ms

FRAMEWORK PyTorch

Dual-Branch Extraction

Spatial Branch

Static Appearance Features

`ResNet-50 (Pre-trained)`

Output: 2048-d Vector

Temporal Branch

Motion Dynamics (16-Frame Clips)

`R(2+1)D Factorization`

Output: 512-d Vector

CONCATENATION FUSION

2560-d Concat Vector

Linear Projection to 512-d

Vision Transformer

Sequence Modeling Layer

`4-Block Encoder`

Training Strategy

- **Clip-Level Supervision**
  Precise temporal annotations over weak labels.
- **Class Weighting**
  Inverse frequency sampling to mitigate normality bias.
- **PyTorch Lightning**
  Structured training loops & hardware scaling.

PERFORMANCE

Top-1 Acc: 74.3%

AUC: 0.89

Implementation

CHAPTER 5: IMPLEMENTATION DETAILS

Inference-Time Safeguards

DECISION ENGINE

Dual-layer verification mechanism to suppress false alarms while maintaining high detection accuracy.

Confidence Filter

THRESHOLD LEVEL

60%

Predictions below **60%** confidence are automatically discarded as background noise.

Temporal Validation

VERIFICATION WINDOW

5 sec

- Ensures anomaly persists consistently across consecutive frames.
- Filters short-term visual noise or momentary irregularities.
- Prevents false alerts triggered by sudden, non-threatening movements.

Input Score → Gate 1: Confidence → Gate 2: Duration → EMERGENCY ALERT

Implementation

Deployment & Robustness

Production Ready

Fault Tolerance

Designed to handle network failures gracefully with localized edge intelligence.

MECHANISM

Offline Buffering

Queues critical alerts locally during disconnects. **Zero data loss** upon network restoration.

Data Privacy Framework

PROCESSING MODE

Ephemeral

Volatile memory only. No disk persistence of raw streams.

RETENTION

Conditional

Clips stored **only if** anomaly alert is triggered.

Deployment

- Containerized Services
- GPU-Accelerated Inference
- Edge / Cloud Hybrid

Operator Dashboard

5.11

Live Feed Monitor

Real-time Stream Analysis

- Role-Differentiated Alerts
- Event Logging & History
- Threshold Configuration

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System Integration

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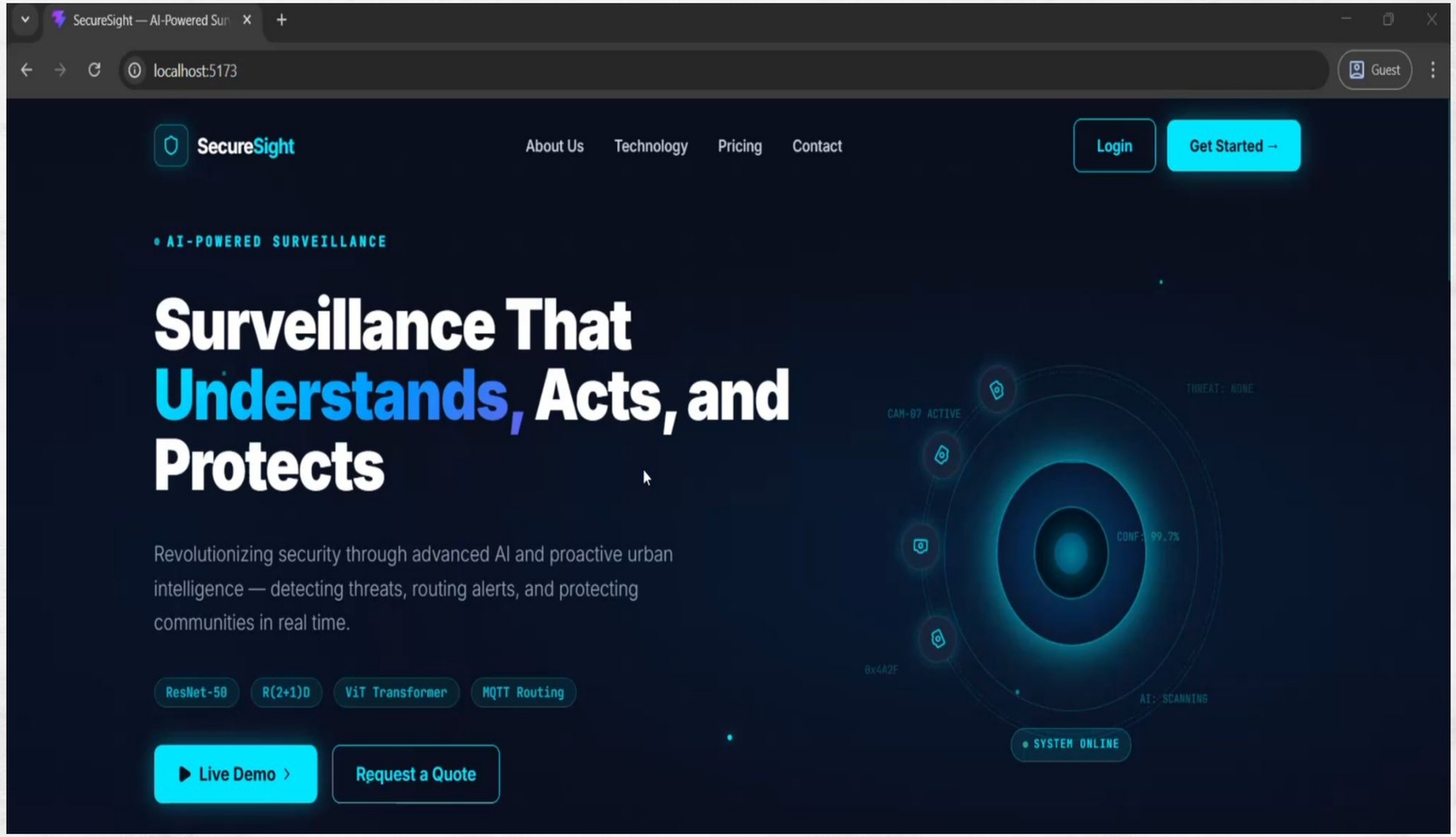
Results

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Web Interface

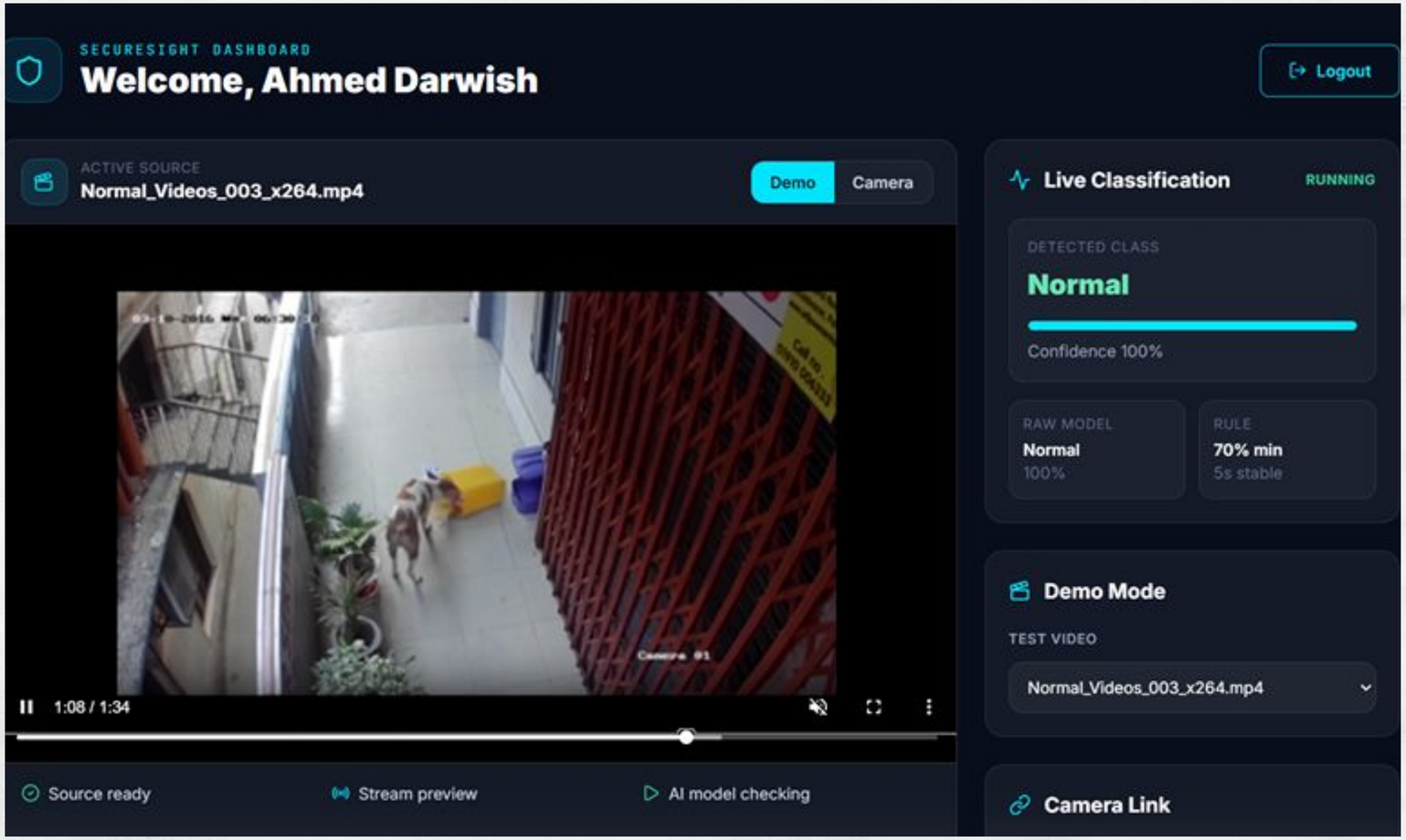
Web Interface

- End-to-end working system
- Real-time video analysis
- Multi-class anomaly detection
- Live alert visualization



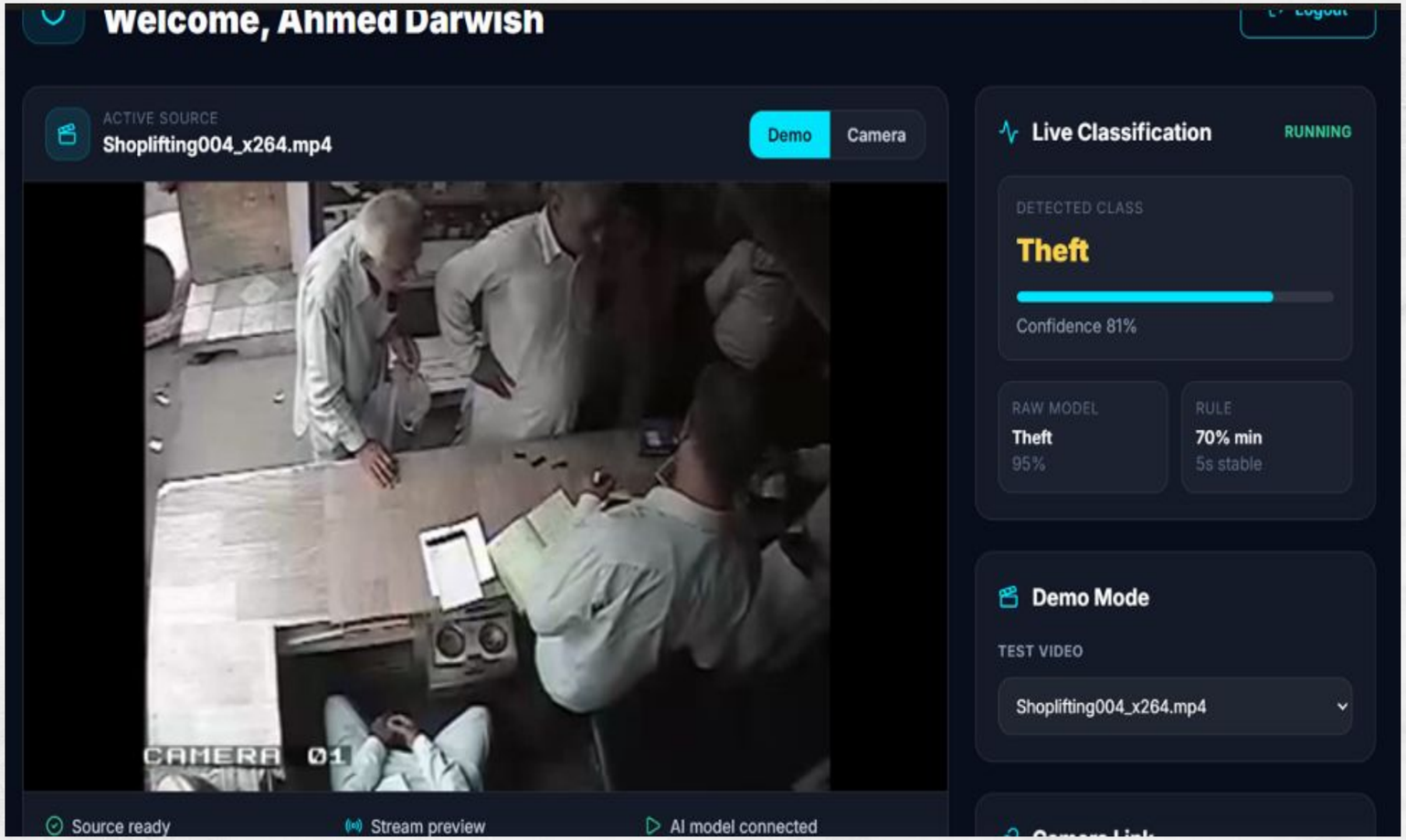
Web Interface

- Input: CCTV / video footage
- Model processes video stream
- Detects anomaly in real-time
- Displays confidence score
- Triggers alerts



Web Interface

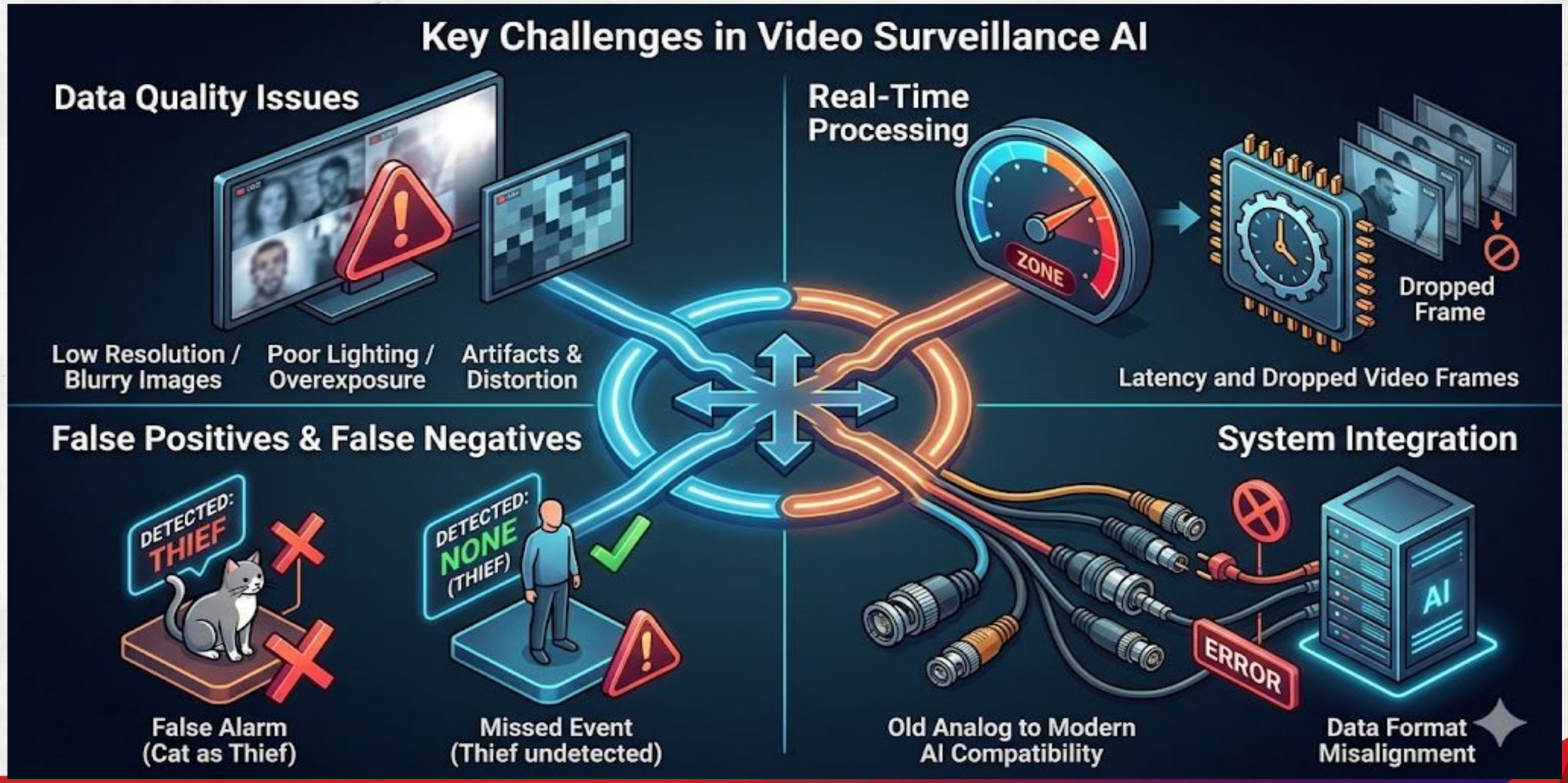
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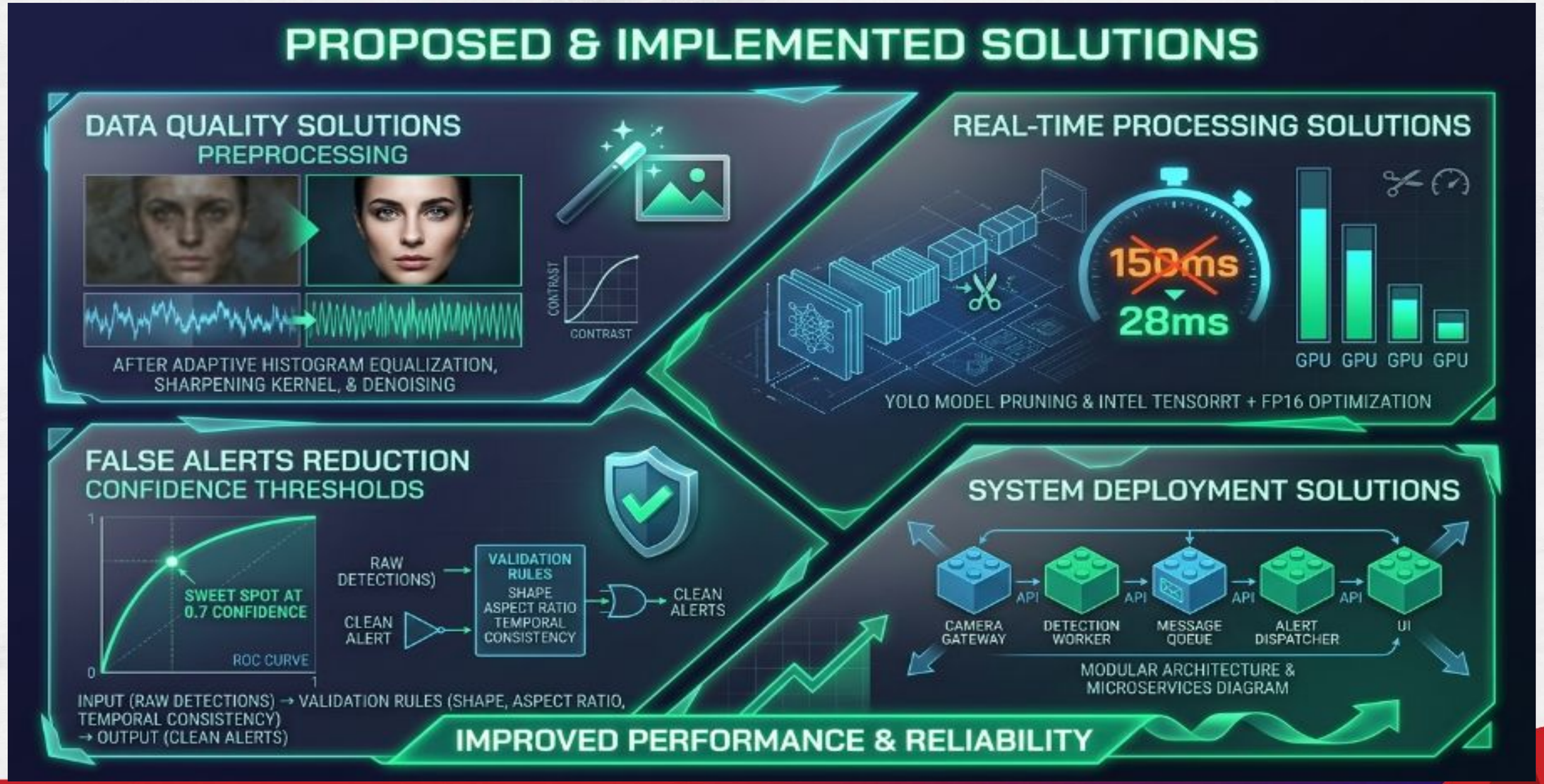
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System Features&Demo

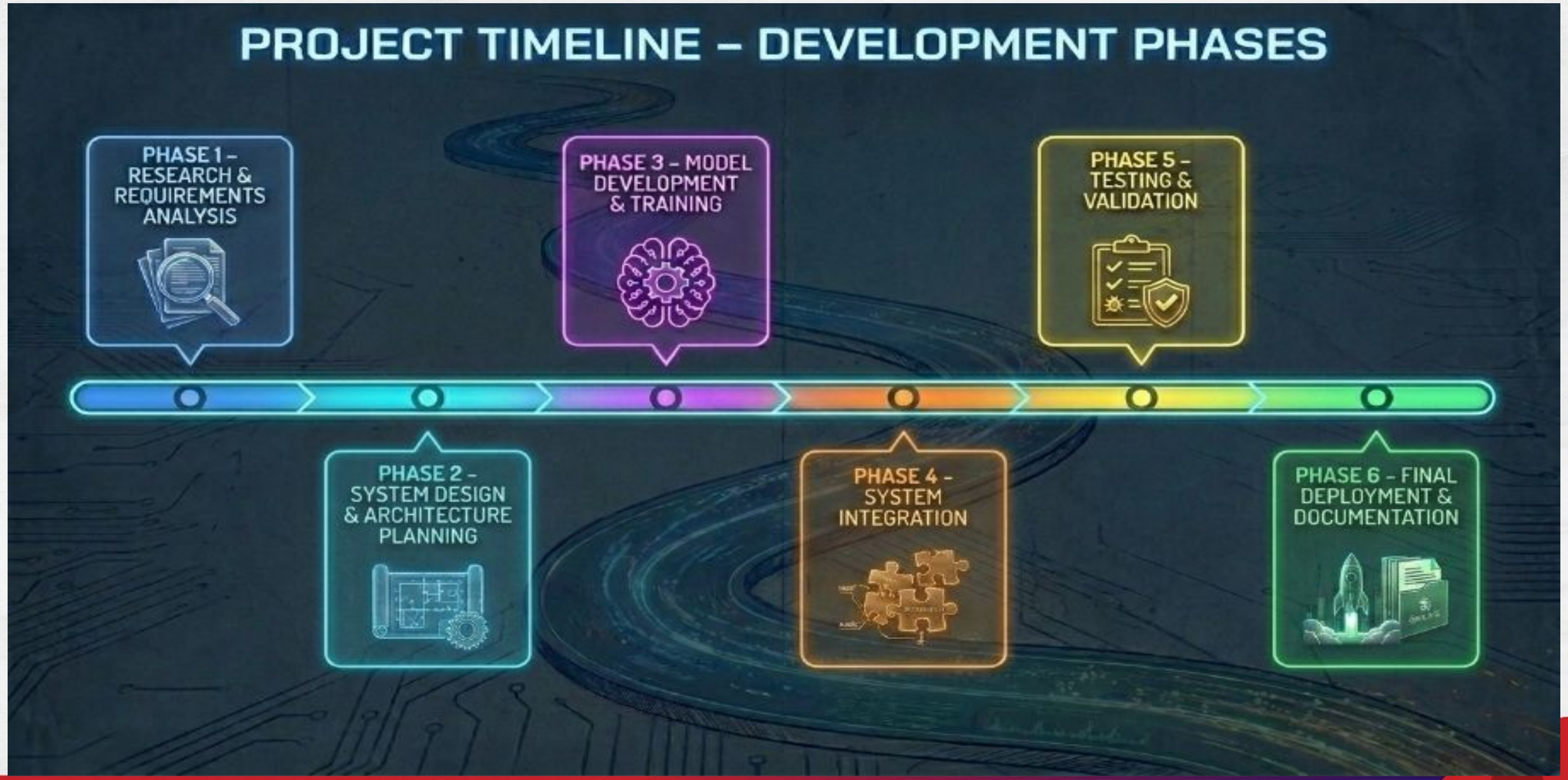
Challenges


Key Challenges in Video Surveillance AI
Data Quality Issues
Low Resolution / Blurry Images
Poor Lighting / Overexposure
Artifacts & Distortion
Real-Time Processing
ZONE
Dropped Frame
Latency and Dropped Video Frames
False Positives & False Negatives
DETECTED: THIEF
False Alarm (Cat as Thief)
DETECTED: NONE (THIEF)
Missed Event (Thief undetected)
System Integration
ERROR
AI
Old Analog to Modern AI Compatibility
Data Format Misalignment

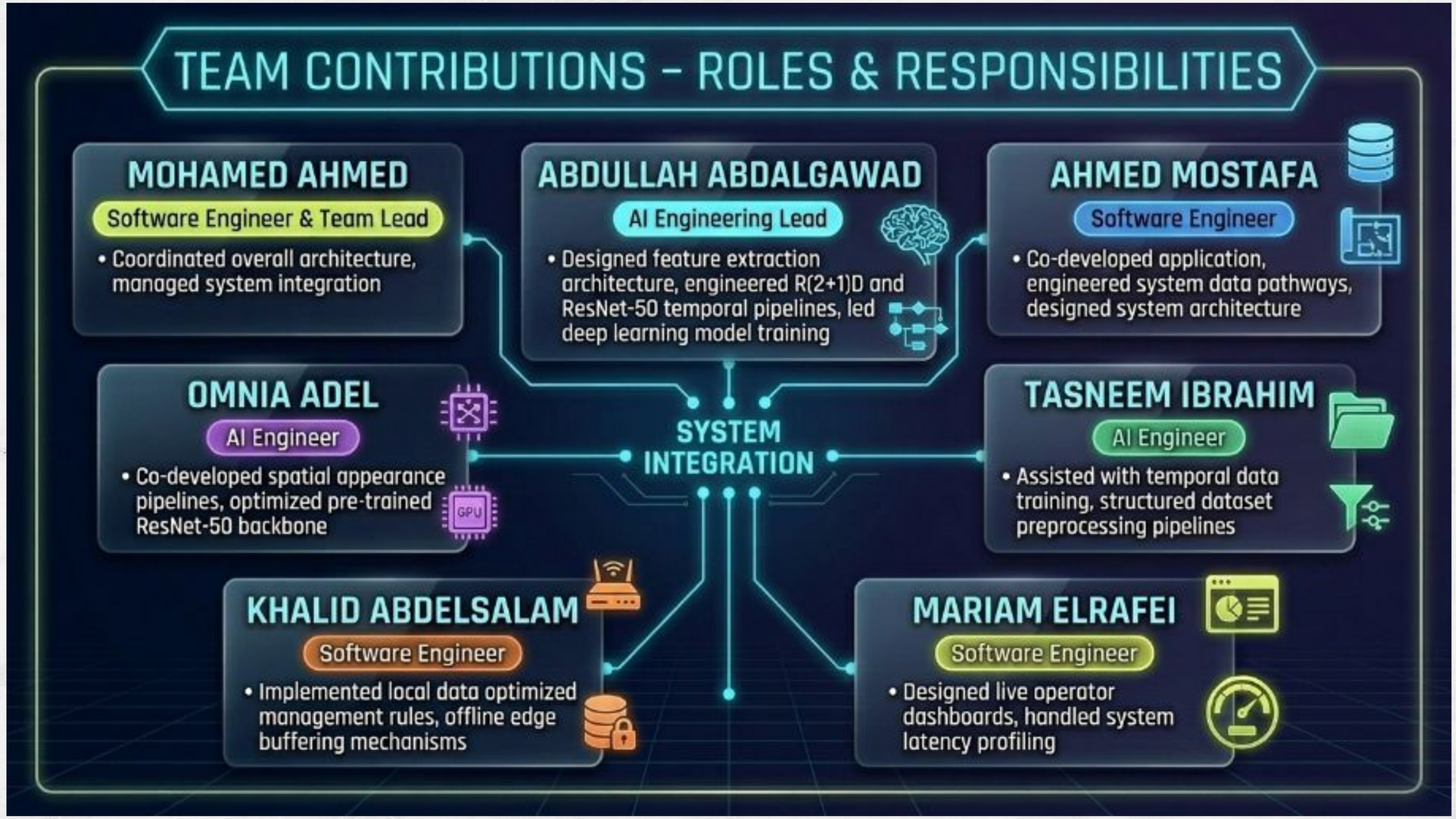
Solutions


PROPOSED & IMPLEMENTED SOLUTIONS
DATA QUALITY SOLUTIONS
PREPROCESSING
CONTRAST
CONTRAST
AFTER ADAPTIVE HISTOGRAM EQUALIZATION, SHARPENING KERNEL, & DENOISING
REAL-TIME PROCESSING SOLUTIONS
150ms
28ms
GPU GPU GPU GPU
YOLO MODEL PRUNING & INTEL TENSORRT + FP16 OPTIMIZATION
FALSE ALERTS REDUCTION
CONFIDENCE THRESHOLDS
1
0
1
SWEET SPOT AT 0.7 CONFIDENCE
ROC CURVE
RAW DETECTIONS)
CLEAN ALERT
VALIDATION RULES
SHAPE
ASPECT RATIO
TEMPORAL CONSISTENCY
CLEAN ALERTS
INPUT (RAW DETECTIONS) → VALIDATION RULES (SHAPE, ASPECT RATIO, TEMPORAL CONSISTENCY)
→ OUTPUT (CLEAN ALERTS)
SYSTEM DEPLOYMENT SOLUTIONS
API
API
API
API
CAMERA GATEWAY
DETECTION WORKER
MESSAGE QOEUE
ALERT DISPATCHER
UI
MODULAR ARCHITECTURE & MICROSERVICES DIAGRAM
IMPROVED PERFORMANCE & RELIABILITY

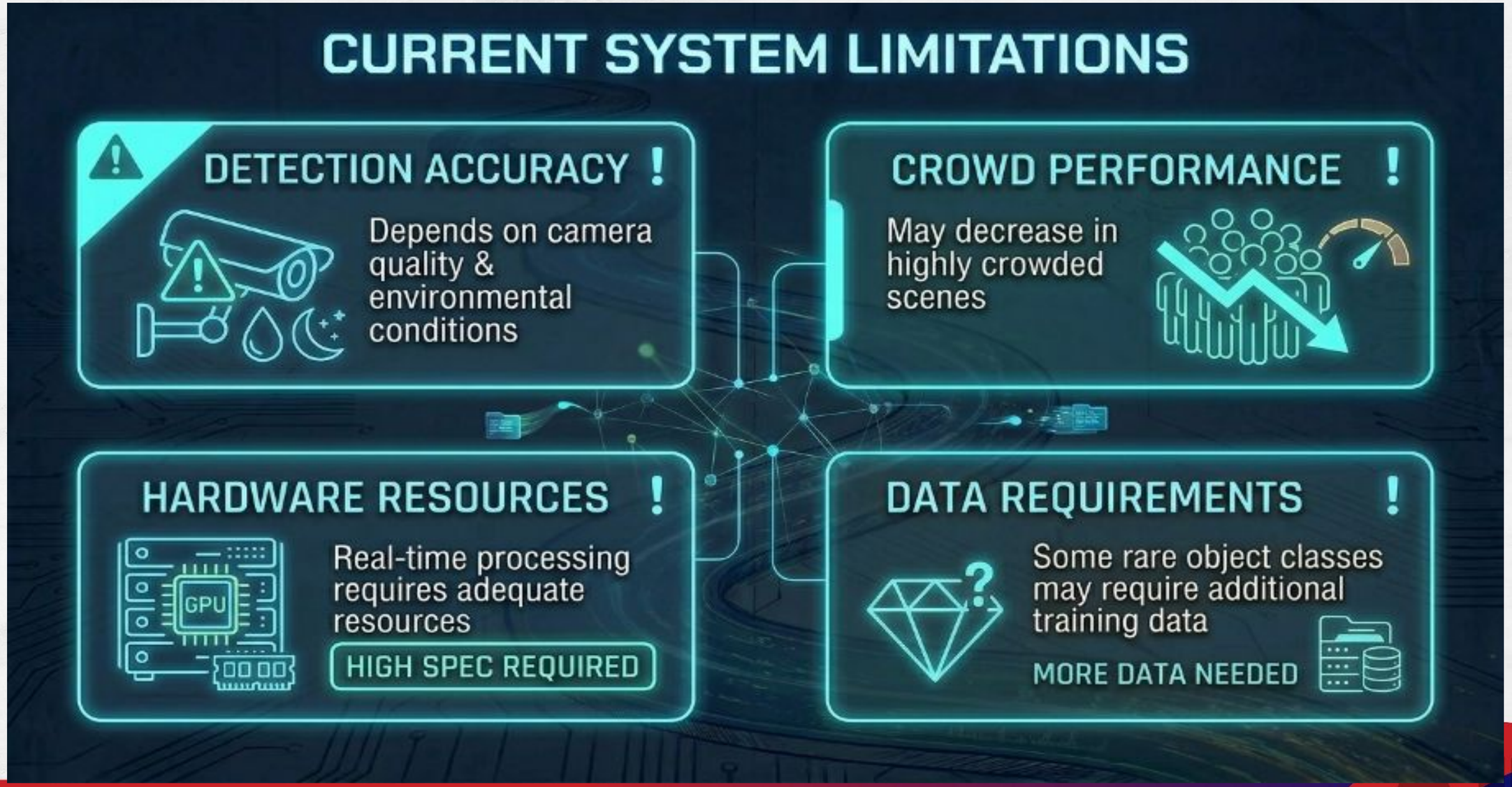
Project Timeline



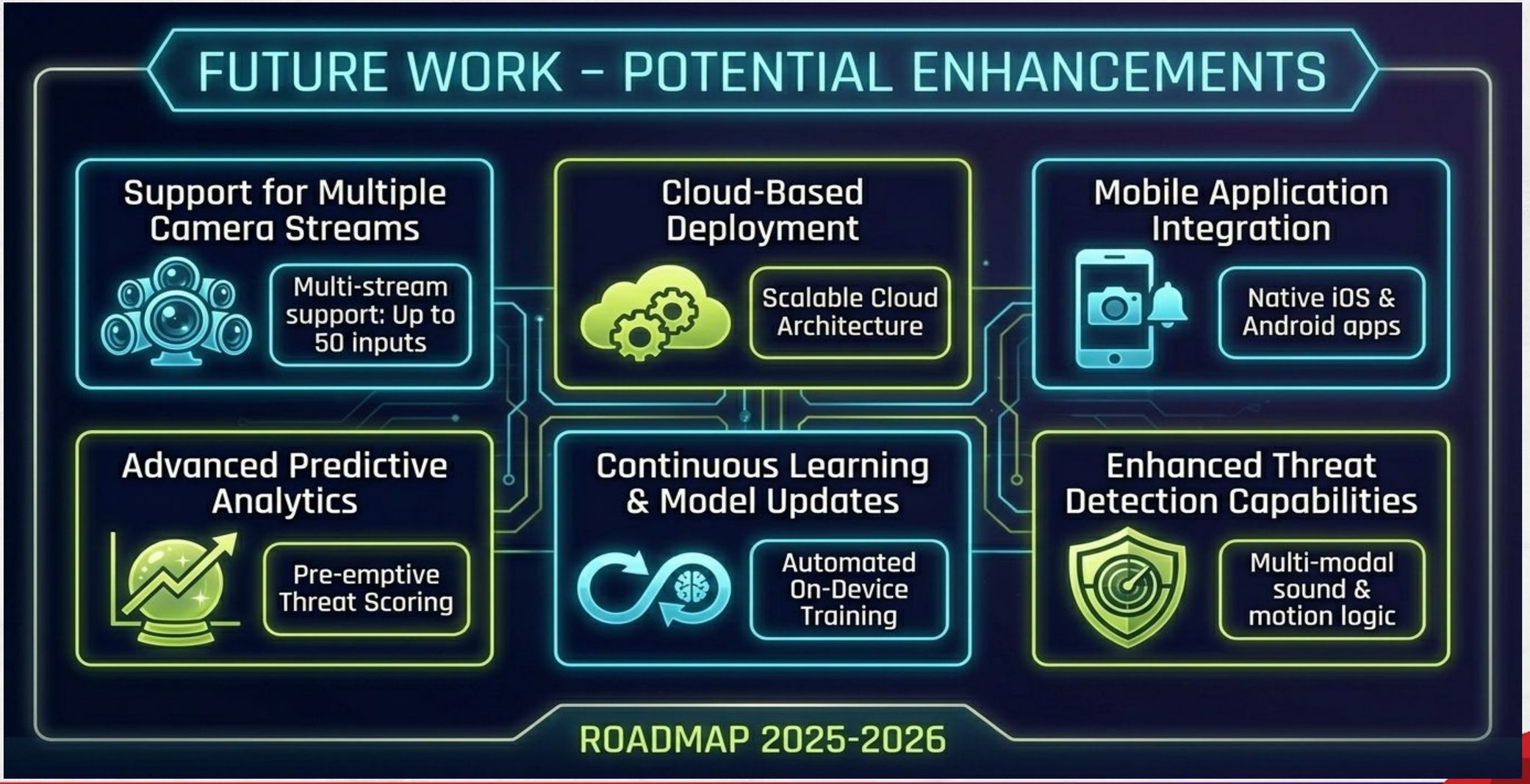
Team Contributions



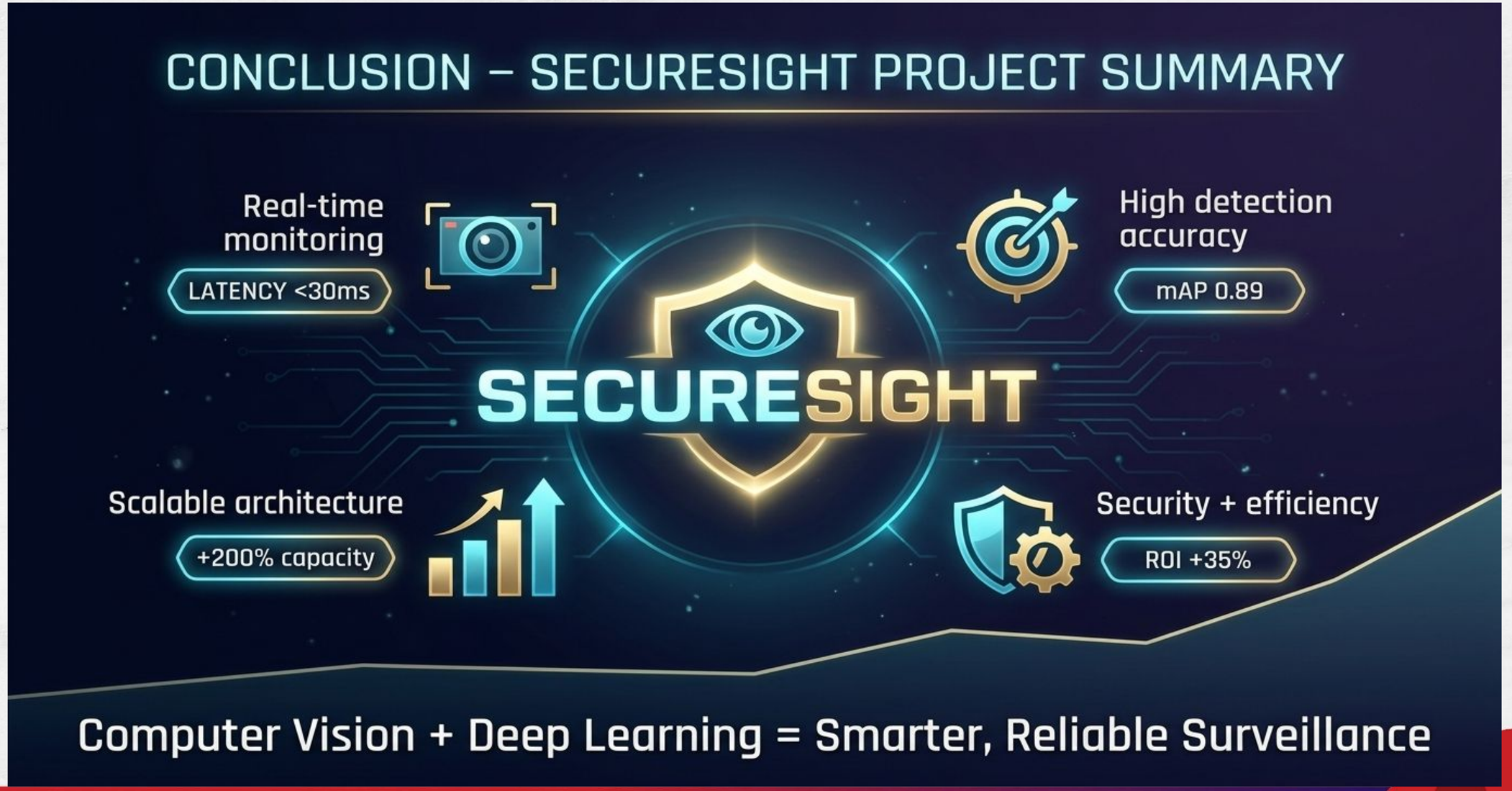
Limitations



Future Work


FUTURE WORK – POTENTIAL ENHANCEMENTS
Support for Multiple Camera Streams
Multi-stream support: Up to 50 inputs
Cloud-Based Deployment
Scalable Cloud Architecture
Mobile Application Integration
Native iOS & Android apps
Advanced Predictive Analytics
Pre-emptive Threat Scoring
Continuous Learning & Model Updates
Automated On-Device Training
Enhanced Threat Detection Capabilities
Multi-modal sound & motion logic
ROADMAP 2025-2026

Conclusion


CONCLUSION – SECURESIGHT PROJECT SUMMARY
Real-time monitoring
LATENCY <30ms
High detection accuracy
mAP 0.89
SECURESIGHT
Scalable architecture
+200% capacity
Security + efficiency
ROI +35%
Computer Vision + Deep Learning = Smarter, Reliable Surveillance

References

- UCF-Crime Dataset – Sultani et al., CVPR 2018
- UCA Annotation Dataset – Yuan et al., arXiv 2023
- ResNet-50 – He et al., CVPR 2016
- I3D – Carreira & Zisserman, CVPR 2017
- R(2+1)D – Tran et al., CVPR 2018
- Vision Transformer (ViT) – Dosovitskiy et al., arXiv 2020
- Graph Convolutional Label Noise Cleaner – Zhong et al., CVPR 2019
- RTFM Method – Tian et al., ICCV 2021

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Q&A